

Notation. Let $a_1, \dots, a_k$ be distinct positive integers with $k \geq 3$ and product $P = \prod_{i=1}^k a_i$. Define $g = \gcd\{P/a_i + 1 \mid 1 \leq i \leq k\}$ and $S = \sum_{i=1}^k a_i$. We are given $S < 3 \cdot 10^6$ and are asked for the maximum possible value of g .

0.0.1 1. Necessary conditions

Since $g \mid (P/a_i + 1)$ for every i , we have

$$\frac{P}{a_i} \equiv -1 \pmod{g} \quad (i = 1, \dots, k).$$

Multiplying by a_i gives $P \equiv -a_i \pmod{g}$, hence all a_i are congruent modulo g . Let r be that common residue, $0 \leq r < g$. From (1) we obtain

$$\frac{P}{a_i} \equiv r^{k-1} \pmod{g},$$

because each of the $k-1$ factors a_j ($j \neq i$) is $\equiv r \pmod{g}$. Consequently

$$\frac{P}{a_i} + 1 \equiv r^{k-1} + 1 \pmod{g}.$$

Since g divides this expression, we get the crucial congruence

$$r^{k-1} + 1 \equiv 0 \pmod{g}.$$

Moreover, if $g \geq 2$ then $r \neq 0$; otherwise all a_i would be multiples of g , making $\frac{P}{a_i} + 1 \equiv 1 \pmod{g}$, impossible for $g > 1$. Thus

$$1 \leq r \leq g-1.$$

Because the a_i are distinct and all $\equiv r \pmod{g}$, any two of them differ by a non-zero multiple of g ; therefore

$$|a_i - a_j| \geq g \quad (i \neq j).$$

This gives a lower bound on the sum S . Order the numbers increasingly: $a_{(1)} < a_{(2)} < \dots < a_{(k)}$. From (4) we obtain

$$a_{(j)} \geq r + (j-1)g \quad (j = 1, \dots, k).$$

Hence

$$S = \sum_{i=1}^k a_i \geq kr + g \frac{k(k-1)}{2}.$$

Together with $S < 3 \cdot 10^6$ we have

$$kr + g \frac{k(k-1)}{2} < 3 \cdot 10^6.$$

0.0.2 2. Bounding g according to k

Case $k \geq 4$ From (6) and $r \geq 1$ we get

$$g \frac{k(k-1)}{2} < 3 \cdot 10^6 \implies g < \frac{6 \cdot 10^6}{k(k-1)}.$$

For $k = 4$ this gives $g < \frac{6 \cdot 10^6}{12} = 500\,000$, so $g \leq 499\,999$. For $k \geq 5$ the bound is even smaller. Hence for **all** $k \geq 4$

$$g \leq 499\,999 < 998\,285.$$

Case $k = 3$ Now $k = 3$ in (2) and (6):

$$r^2 + 1 \equiv 0 \pmod{g}, \quad 3r + 3g < 3 \cdot 10^6 \implies r + g \leq 999\,999.$$

Write $r^2 + 1 = mg$ with $m \in \mathbb{N}$. Because $r \leq g - 1$, we have $m \geq 1$. Our aim is to show $g \leq 998\,285$.

Assume, for contradiction, that $g \geq 998\,286$.

From $r^2 + 1 \geq g$ we obtain $r^2 \geq g - 1 \geq 998\,285$; therefore

$$r \geq 1000.$$

Now $r + g \leq 999\,999$ together with $g \geq 998\,286$ yields

$$r \leq 999\,999 - g \leq 999\,999 - 998\,286 = 1\,713.$$

Thus $1000 \leq r \leq 1\,713$.

Because $r^2 + 1 = mg$ and $g \geq 998\,286$, we have

$$m = \frac{r^2 + 1}{g} \leq \frac{1\,713^2 + 1}{998\,286} \approx 2.94.$$

Hence m can only be 1 or 2.

- **If $m = 1$** , then $g = r^2 + 1$. Then

$$r + g = r + r^2 + 1 = r^2 + r + 1 \geq 1000^2 + 1000 + 1 = 1\,001\,001 > 999\,999,$$

contradicting (8).

- If $m = 2$, then $g = \frac{r^2 + 1}{2}$. Now

$$r + g = r + \frac{r^2 + 1}{2} = \frac{r^2 + 2r + 1}{2} = \frac{(r + 1)^2}{2}.$$

Condition (8) becomes $\frac{(r + 1)^2}{2} \leq 999\,999$, i.e. $(r + 1)^2 \leq 1\,999\,998$. Hence $r + 1 \leq 1414$ and therefore $r \leq 1413$.

On the other hand, $g \geq 998\,286$ gives

$$\frac{r^2 + 1}{2} \geq 998\,286 \implies r^2 + 1 \geq 1\,996\,572 \implies r^2 \geq 1\,996\,571.$$

Since $1412^2 = 1\,993\,744 < 1\,996\,571$ and $1413^2 = 1\,996\,569$, we must have $r \geq 1413$.

Combining $r \leq 1413$ and $r \geq 1413$ forces $r = 1413$, but then

$$g = \frac{1413^2 + 1}{2} = \frac{1\,996\,569 + 1}{2} = 998\,285,$$

which is **strictly smaller** than the assumed $g \geq 998\,286$. Contradiction.

Thus our assumption $g \geq 998\,286$ is impossible; therefore

$$g \leq 998\,285 \quad \text{for } k = 3.$$

0.0.3 3. Combining the cases

From (7) we have $g \leq 499\,999$ for $k \geq 4$, and from (11) we have $g \leq 998\,285$ for $k = 3$. Hence **for every admissible choice of a_i and $k \geq 3$**

$$g \leq 998\,285.$$

0.0.4 4. Achieving the bound

It remains to exhibit a concrete tuple for which the gcd reaches 998 285.

Take $k = 3$, $r = 1413$ and $g = 998\,285$. Choose

$$a_1 = 1413, \quad a_2 = 1413 + 998\,285 = 999\,698, \quad a_3 = 1413 + 2 \cdot 998\,285 = 1\,997\,983.$$

These numbers are distinct positive integers. Their sum is

$$1413 + 999\,698 + 1\,997\,983 = 2\,999\,094 < 3 \cdot 10^6.$$

Now verify that 998 285 divides each $P/a_i + 1$. Because each $a_j \equiv r \pmod{g}$ and

$$r^2 + 1 = 1413^2 + 1 = 1\,996\,570 = 2 \cdot 998\,285 \equiv 0 \pmod{g},$$

we have $r^2 + 1 \equiv 0 \pmod{g}$. For $i = 1$,

$$\frac{P}{a_1} = a_2 a_3 \equiv r^2 \pmod{g}, \quad \frac{P}{a_1} + 1 \equiv r^2 + 1 \equiv 0 \pmod{g}.$$

The same computation holds for $i = 2, 3$. Hence $998\,285 \mid (\frac{P}{a_i} + 1)$ for all i .

Consequently the greatest common divisor g of these three numbers is a multiple of 998 285. Because we have already proved that $g \leq 998\,285$ for **any** possible choice, it follows that for this particular choice g equals exactly 998 285.

0.0.5 5. Answer

998 285
